

AP Precalculus: 2026–2027

Unit 1: Polynomial and Rational Functions Lesson 1.14: Function Model Construction and Application

Primary Objective: Students will construct polynomial, piecewise, and rational function models from contextual descriptions using restrictions and inverse-proportion relationships, and apply those models to predict values and compute average rates of change with appropriate units. *Skills: MP 1.C, MP 3.B*

Priority Ideas & Skills

AP Skills

MP 1.C — Construct new functions using transformations, compositions, or regression.

MP 3.B — Apply numerical results in context using appropriate units.

Key Understandings

- A model is built from a contextual restriction by expressing one variable in terms of another, then substituting into the objective formula. (EK 1.14.A.1, A.2)
- Inversely proportional quantities $F \propto 1/d^n$ are modeled by rational functions $r(d) = k/d^n$, with domain $d > 0$. (EK 1.14.B.1)
- The average rate of change $\frac{f(b)-f(a)}{b-a}$ has units equal to output units divided by input units and describes the model's behavior over an interval. (EK 1.14.C.1)

Vocabulary, Concepts, & Theorems

function model	A function constructed to represent a real-world scenario; used to predict values.
restriction	A constraint on input values derived from the context (e.g., length must be positive).
regression	A technology-assisted technique for fitting a curve of a given type to a data set.
average rate of change	$\frac{f(b) - f(a)}{b - a}$; slope of the secant line; units = output/input.
inverse proportion	A relationship $y = k/x^n$; as x increases, y decreases.
domain restriction	The set of allowable inputs dictated by the contextual meaning of the variable.
piecewise model	A function defined by different formulas on different intervals of its domain.

Activate Prior Knowledge & Spiral Review

Spiral review (warm-up) covers: (1) computing average rate of change from a table with correct units (from 1.1/1.4); (2) evaluating a polynomial at a given input (from 1.2); (3) recognizing inverse-square language as a rational model (bridges to 1.14.B).

Hook

“A farmer has 80 feet of fencing and wants the biggest pen possible. One side is against the barn — no fencing needed there. Can a function tell us the answer?”

Write the scenario on the board. Have students predict: would the optimal shape be a square? Why or why not? Take 3–4 verbal responses. Then say: “Let’s use a function model to find out exactly.”

Lesson

Part 1 (15 min): Model from Restrictions (Notes Sec. 1).

1. Label the pen: two widths w , one length ℓ , barn on fourth side.
2. Constraint: $2w + \ell = 80 \Rightarrow \ell = 80 - 2w$.
3. Objective: $A = w\ell \Rightarrow A(w) = -2w^2 + 80w$.
4. Domain: $0 < w < 40$.
5. Vertex: $w = 20$ ft, $A(20) = 800$ ft².

Pose: **“Is this a square pen?”** ($\ell = 40 \neq w = 20$, so no.)

Part 2 (10 min): Rational Model (Notes Sec. 2).

Two charged objects: force $\propto 1/d^2$, so $r(d) = k/d^2$.

- Domain: $d > 0$ (physical distance must be positive).
- As $d \rightarrow 0^+$: force $\rightarrow \infty$. As $d \rightarrow \infty$: force $\rightarrow 0$.
- Complete table with $k = 900$: $d = 1, 3, 5, 10$.

Part 3 (10 min): Apply the Model (Notes Sec. 3).

Return to $A(w) = -2w^2 + 80w$:

- Evaluate $A(10) = 600$ ft² and $A(20) = 800$ ft².
- ARC on $[10, 20]$: $\frac{800-600}{10} = 20$ ft²/ft.
- Interpret: each foot of additional width yields 20 ft² more area, on average, between $w = 10$ and $w = 20$.

Active Monitoring

- After Part 1 — cold call: **“Why is the domain $0 < w < 40$, not $w > 0$?”** (If $w \geq 40$, $\ell \leq 0$; no pen possible.)
- After Part 2 — cold call: **“Why can’t $d = 0$ in your rational model?”** (Division by zero; also physically impossible.)
- After Part 3 — cold call: **“What units does the average rate of change have?”** (ft²/ft. Push students to read units as “square feet per foot of width.”)
- Watch for: students computing ARC without including units; students plugging into the original unsimplified expression rather than $A(w) = -2w^2 + 80w$.

Group Work & Differentiation

Tier R — Remediate

- Area model pre-set up: evaluate $A(w) = -2w^2 + 60w$ at four widths.
- Compute ARC on $[5, 15]$.
- Identify the approximate maximum from the table.

Tier A — Apply

- Write rational model $C(d) = 2000/d^2$ from description.
- State domain restriction with explanation.
- Complete table; compute ARC on $[2, 4]$ with units.

Tier E — Extend

- Construct quadratic model by solving 3×3 system from data points.
- State modeling assumptions.
- Predict at $x = 4$ (extrapolation) and comment.

Individual Work & Assessment

Exit Ticket (3 questions, 1 page):

1. Evaluate $f(x) = -2x^2 + 12x$ at $x = 3$ and $x = 5$; compute ARC on $[3, 5]$ with units.
2. State domain restriction for $P(d) = 50/d^2$ in context.
3. Interpret a negative ARC in the pen design context.

Collect exit tickets as students leave. Sort into three piles:

- **Missing units only** — quick fix in warm-up next class.
- **Incorrect ARC computation** — small-group pull during Lesson 1.14 practice.
- **All correct** — ready for Unit 1 review.

Reinforcement & Extension

Homework (6 problems):

- (1) Model from restrictions (two-sided garden): derive $A(w)$, domain, vertex.
- (2) Rational gravitational model: domain, end behavior, inverse-square ratio.
- (3) Projectile $s(t) = -16t^2 + 64t + 5$: evaluate, ARC, interpret zero ARC.
- (4) Tank draining: interpret sign of ARC; estimate output from average.
- (5) Piecewise parking-garage model: write, evaluate, classify.
- (6) AP-style MC on ARC of a revenue model.

Preview — Unit 1 Sample Test: Lesson 1.14 is the final lesson of Unit 1. After returning homework, distribute the Unit 1 Sample Test for in-class or at-home review. Topics covered on the test span 1.1–1.14: polynomial behavior, rational functions (VA, holes, end behavior), and function model construction and application.